

LP2 (P4 – N-QUEEN BACKTRACKING)

Implement a solution for a Constraint Satisfaction Problem using Branch and Bound and Backtracking for n-queens problem or a graph colouring problem

```
def issafe(arr, x, y, n):
```

```
    for row in range(x):
```

```
        if arr[row][y] == 1:
```

```
            return False
```

```
    row, col = x, y
```

```
    while row >= 0 and col >= 0:
```

```
        if arr[row][col] == 1:
```

```
            return False
```

```
        row -= 1
```

```
        col -= 1
```

```
    row, col = x, y
```

```
    while row >= 0 and col < n:
```

```
        if arr[row][col] == 1:
```

```
            return False
```

```
        row -= 1
```

```
        col += 1
```

```
    return True
```

```

def nQueen(arr, x, n):
    if x >= n:
        return True

    for col in range(n):
        if issafe(arr, x, col, n):
            arr[x][col] = 1
            if nQueen(arr, x+1, n):
                return True
            arr[x][col] = 0

    return False

n = int(input("Enter number of Queens: "))
arr = [[0]*n for i in range(n)]
if nQueen(arr, 0, n):
    print("\nSolution:\n")
    for i in arr:
        print(*i)
else:
    print("No Solution")

```

Viva Questions & Answers — N-Queens Problem

1. What is N-Queens Problem?

N-Queens problem is placing N queens on an N×N chessboard such that no two queens attack each other.

2. Which algorithm did you use?

Backtracking algorithm.

3. What is backtracking?

Backtracking is a recursive technique that tries solutions and removes wrong choices when constraints fail.

4. What is Constraint Satisfaction Problem (CSP)?

A problem where solutions must satisfy given constraints.

Example:

- N-Queens
 - Graph coloring
 - Sudoku
-

5. Why is N-Queens called a CSP?

Because queens must satisfy constraints:

- No same row
 - No same column
 - No same diagonal
-

6. What are the constraints in N-Queens?

Queens cannot attack each other:

- Horizontally
 - Vertically
 - Diagonally
-

7. What is pruning?

Pruning means stopping unnecessary or invalid search paths early.

8. Where is pruning used in your code?

In:

```
if issafe(arr, x, col, n):
```

Unsafe positions are rejected immediately.

9. What is Branch and Bound?

Branch and Bound is an optimization technique that prunes non-promising states to reduce searching.

10. Does your program use Branch and Bound?

The program mainly uses backtracking with pruning similar to Branch and Bound.

11. What is recursion?

A function calling itself repeatedly.

12. Which function is recursive in your code?

nQueen()

13. What is the purpose of issafe()??

To check whether a queen placement is safe.

14. What checks are performed in issafe()??

- Column check
 - Left diagonal check
 - Right diagonal check
-

15. Why row checking is not needed?

Because only one queen is placed per row.

16. Why do we backtrack?

To remove wrong queen placements and try another possibility.

17. Which line performs backtracking?

arr[x][col] = 0

18. What happens when solution is found?

if $x \geq n$:

 return True

All queens are successfully placed.

19. What is the base condition?

if $x \geq n$

20. Why do we use recursion in N-Queens?

To solve row-by-row repeatedly.

21. What is the time complexity of N-Queens?

Approximately:

$O(N!)$

22. What is the space complexity?

$O(N^2)$

for chessboard storage.

23. Why is N-Queens difficult?

Because possibilities increase rapidly as N increases.

24. What happens if no solution exists?

Program prints:

No Solution

25. What does `print(*i)` do?

Prints list elements in a row format.

26. What does value 1 represent?

Queen placed.

27. What does value 0 represent?

Empty position.

28. Why are loops used in issafe()?

To check attacks from previously placed queens.

29. Why only upper rows are checked?

Because queens are placed row by row from top to bottom.

30. Can multiple solutions exist?

Yes, N-Queens may have multiple valid solutions.

31. Why does your program print only one solution?

Because it returns True after first successful solution.

32. How can you print all solutions?

Remove immediate return True and continue recursion.

33. Applications of Backtracking?

- Sudoku
 - Maze solving
 - Graph coloring
 - N-Queens
-

34. Difference between Backtracking and Brute Force?**Backtracking****Brute Force**

Prunes invalid paths

Checks all possibilities

Faster

Slower

Optimized

Non-optimized

35. Why is backtracking efficient?

Because invalid states are rejected early using pruning.

36. Which data structure is used?

2D List (Matrix).

37. What is diagonal attack condition?

Queens attack diagonally if row and column differences are equal.

38. Why is N-Queens important in AI?

It demonstrates:

- Constraint satisfaction
 - Search techniques
 - Backtracking
 - State space search
-

39. What is state space?

Set of all possible configurations/states.

40. One-line summary of your program?

“This program solves the N-Queens Constraint Satisfaction Problem using backtracking and pruning techniques.”